

PMT Notes

for week 2

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This set of notes is written for the course 40018: Discrete Mathematics, Logic & Reasoning, taught in Imperial College London 2024-2025 by Dr Steffen van Bakel.

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§1. An additional question

This question is inspired by isomorphisms in category theory when I am learning this subject recently. Category theory is an interesting subject that connects with discrete mathematics and logic really tightly. Go for this if you are interested in theoretical foundations of computer science and Haskell!

Problem 1.1.

Let C be a set. Let A, B be two sets satisfying $A, B \in C$.

(so imagine something like $C = \{\{1, 2\}, \{3, 4, 5\}\}$, and $A = \{1, 2\}$)

For a relation R between sets A and B , we define

- R is iso $\triangleq \exists (S \subseteq B \times A). (S \circ R = \text{id}_B) \wedge (R \circ S = \text{id}_A)$.
- In other words, R is iso $\triangleq$ if there exists another relation S between set B and A , such that $S \circ R = \text{id}_B$, and $R \circ S = \text{id}_A$.

we denote:

- $A \cong B \triangleq$ if there exists some $R \subseteq A \times B$, such that R is iso.

Show that $\cong$ is an equivalence relation on the elements in C . (i.e., $\cong$ is reflexive, symmetric, and transitive),